EU AI Act Product Classification Lab

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LAB | Classify your product

Role: AI consultant

First-pass consulting assessment, not legal advice

Contents

Submission map

This file contains:

1. A private answer key with the intended classifications.
2. Four partner-facing client briefs with the category labels removed.
3. A consulting approval pack covering classification, architecture, roles, controls, and launch decisions.
4. A simulated debrief and final sign-off note.
5. A mini implementation roadmap for the strongest high-risk case.

Part A: Private answer key

Do not share this section with the reviewing partner until after the classification exercise.

Case	Intended category	Reason
1	Prohibited	The system infers workers' emotions from facial and vocal signals and uses the result for workplace evaluation, which falls within Article 5 unless a genuine medical or safety exception applies.
2	High-risk	The system is a safety component used to manage electricity supply and grid switching, bringing it within Annex III point 2 on critical infrastructure.
3	Limited risk with transparency obligations	The system interacts directly with citizens as a chatbot but does not determine grant eligibility, triggering Article 50 disclosure at minimum.
4	Minimal risk	The system optimises solar-panel cleaning schedules and does not make safety-critical decisions or decisions concerning natural persons.

Part B: Partner-facing client briefs

Case 1

A European logistics company wants to understand why some warehouse teams show declining productivity during daily video briefings. The proposed tool would analyse webcam images, facial movements, voice tone, and speaking patterns to estimate each employee's engagement, stress, and motivation. A weekly score would be shown to line managers and could influence coaching, shift allocation, and performance reviews. Employees would be informed that the calls are recorded, but they could not opt out. Managers would see the score before meeting the employee, although they could manually change their final evaluation.

Case 2

A regional electricity distribution operator wants to reduce outages caused by congestion and distributed renewable generation. The proposed system would combine SCADA measurements, smart-meter aggregates, equipment status, weather forecasts, and historical faults to predict overloads and recommend switching actions. During severe congestion, the current design would allow the system to issue switching commands automatically and notify the control-room engineer afterwards. The solution would be delivered under the consultancy's product name and operated by the utility.

Case 3

A municipality wants a multilingual online assistant that answers questions about household energy-renovation grants. It would retrieve information from approved municipal and EU guidance, explain application steps, and direct users to relevant forms. It would not calculate eligibility, rank applications, or submit decisions. Citizens could request transfer to a human adviser. The municipality wants the interface to look like its ordinary help service and has not yet decided whether to identify it as an AI assistant.

Case 4

A solar-farm operator wants to reduce unnecessary panel-cleaning visits. The system would use weather forecasts, irradiance, soiling sensors, historical output, water availability, and maintenance costs to recommend cleaning dates. The output would be a weekly maintenance schedule reviewed by the operations manager. It would not control electrical protection, grid switching, worker evaluation, or access to services. Employee names would not be included in the optimisation data.

Part C: Consulting response and approval pack

Executive summary

The four proposals span the required AI Act outcomes. Case 1 cannot proceed because workplace emotion inference is prohibited. Case 2 can proceed only through the high-risk compliance pathway and should be redesigned so a qualified control-room operator authorises consequential switching. Case 3 can proceed with an explicit AI disclosure, bounded retrieval, and human escalation. Case 4 can proceed under proportionate governance because it is a minimal-risk operational optimisation tool.

Case	First-pass category	Decision
1	Prohibited practice	Deny and redesign
2	High-risk critical-infrastructure AI	Approve with controls
3	Transparency or limited risk	Approve with controls
4	Minimal risk	Approve

Case 1: Workforce engagement analyser

First-pass classification

Prohibited under Article 5 because it infers emotions of workers in a workplace and uses those inferences for evaluation. The safety or medical exception does not apply to productivity, motivation, shift allocation, or performance management.

Proposed architecture as requested

Video briefing -> facial and vocal analysis -> inferred emotional labels -> employee score -> manager decision.

Role map

- System provider: The vendor placing the finished product on the market under its name.
- Deployer: The logistics company.
- Affected persons: Employees.
- Consultant: Adviser or integrator unless the consultancy supplies the branded finished system.

Decision

Deny and redesign. Disclosure, consent language, or a human override does not make the proposed emotion-inference practice lawful.

Lawful alternative

Use voluntary pulse surveys, direct employee feedback, workload measures, incident reports, absenteeism trends, and team-level operational indicators without inferring emotions from biometric signals. Managers should review context and speak with employees before taking any employment action. GDPR, employment, worker-consultation, and discrimination requirements must be reviewed separately.

Case 2: Grid congestion and switching system

First-pass classification

High-risk under Article 6 and Annex III point 2 because the AI system is intended to function as a safety component in managing electricity supply. Incorrect switching could endanger grid stability, health, safety, or essential services.

Controlled architecture

SCADA, meter aggregates, asset status, weather, and fault history -> forecasting and contingency model -> risk-ranked switching recommendation -> qualified operator review and authorisation -> execution through the existing control system -> immutable event log, performance monitoring, and incident workflow.

Automatic emergency action should be limited to predetermined, independently validated protection functions with a documented safety justification. The AI recommendation must not silently replace the authorised operator.

Role map

- Provider: The consultancy if it supplies the completed system under its own product name.
- Deployer: The regional distribution operator.
- Component vendors: Cloud, forecasting, model, cybersecurity, and SCADA-interface suppliers.
- Affected parties: Electricity customers, critical facilities, employees, and connected generators.

Required controls

- Lifecycle risk-management system.
- Data-quality and governance controls.
- Technical documentation and automatic logging.
- Clear operating instructions and limitations.
- Effective human oversight and override capability.
- Accuracy, robustness, fail-safe behaviour, and cybersecurity testing.
- Quality-management system and conformity assessment.
- EU declaration of conformity, CE marking, and registration when applicable.
- Post-market monitoring and Article 73 incident reporting.
- AI literacy and role-specific operator training.
- NIS2, GDPR, energy-sector, and critical-infrastructure review.

Article 27 does not impose a FRIA for Annex III point 2 critical-infrastructure systems, but a broader rights, safety, cybersecurity, and data-protection assessment remains advisable.

Decision

Approve with controls. Launch is conditional on redesigning the human-oversight point, completing the high-risk evidence package, validating fail-safe operation, and obtaining legal confirmation of the final classification and conformity route.

Case 3: Energy-grant information chatbot

First-pass classification

Limited or transparency risk. It interacts directly with natural persons but does not assess eligibility or determine access to a public benefit. Article 50 disclosure applies at minimum.

Controlled architecture

Citizen question -> retrieval from an approved and version-controlled document collection -> answer with citations and uncertainty warning -> refusal when the evidence is insufficient -> transfer to a human adviser -> interaction and content-version record.

Role map

- GPAI provider: The underlying model supplier, if a GPAI model is used.
- System provider: The entity that builds and offers the completed chatbot under its name.
- Deployer: The municipality.
- Affected persons: Citizens seeking grant information.

Required controls

- Clear notice that the user is interacting with AI unless this is already obvious.
- No automated eligibility, ranking, rejection, or approval.
- Human escalation for individual or legally consequential questions.
- Approved sources, citations, update control, and hallucination testing.
- Privacy notice, data minimisation, retention rules, and security controls.
- Accessibility and multilingual quality assurance.
- AI-literacy measures for staff operating and supervising the service.

Decision

Approve with controls. Approval depends on visible AI disclosure, a strict informational boundary, evidence-based retrieval, and human referral.

Case 4: Solar-panel cleaning optimiser

First-pass classification

Minimal risk because the system recommends maintenance timing and does not perform a prohibited function, control a safety-critical electricity component, or make decisions about individuals.

Architecture

Weather, irradiance, soiling, energy output, water, and maintenance-cost data -> optimisation model -> proposed weekly cleaning plan -> operations-manager review -> maintenance schedule and performance record.

Role map

- Provider: The optimisation-product supplier.
- Deployer: The solar-farm operator.
- Vendor: Weather-data or cloud-service provider where applicable.

Proportionate controls

- Confirm that the intended purpose remains maintenance optimisation.
- Validate forecast accuracy and monitor performance drift.
- Maintain access controls, basic logs, backup, and change records.
- Apply AI-literacy measures to relevant staff.
- Reassess classification if the system later controls electrical protection, evaluates workers, or issues autonomous safety-critical commands.
- Apply GDPR if later versions introduce worker or location data.

Decision

Approve. Routine governance and change control are sufficient for the proposed use.

Part D: Simulated client debrief and sign-off

The client discussion clarified that Case 2 was initially intended to execute switching automatically. The recommendation therefore changed from conditional approval of the original design to approval only after placing a competent operator before the consequential switching action. Case 3 was also clarified as an information service rather than an eligibility tool, confirming the transparency classification. The private answer key and consultant classifications matched for all four cases.

Final sign-off status:

- Case 1: Rejected as proposed; redesign requested.
- Case 2: Conditionally approved for controlled development and evidence preparation, not unrestricted production launch.
- Case 3: Approved subject to transparency and safe-use controls.
- Case 4: Approved with routine governance.

Part E: Mini implementation roadmap for Case 2

Provider before market placement

1. Confirm the intended purpose, Annex III point 2 classification, system boundary, and role allocation.
2. Establish quality and risk-management systems.
3. Document data sources, quality, representativeness, failure modes, and validation.
4. Implement logging, instructions, human oversight, accuracy, robustness, cybersecurity, and fail-safe controls.
5. Complete the applicable conformity assessment and prepare the declaration, marking, and registration.
6. Establish post-market monitoring, change control, and serious-incident reporting.

Deployer before first use

1. Verify registration and provider documentation.
2. Configure the system within the validated intended purpose.
3. Appoint and train authorised control-room personnel.
4. Validate local data, integration, fallback operation, and override authority.
5. Establish log retention, monitoring, escalation, and suspension procedures.
6. Complete sector, cybersecurity, safety, and data-protection assessments.

Evidence requested from vendors

- System and model cards, intended-purpose statement, and limitations.
- Data provenance, quality, validation, and error-analysis reports.
- Accuracy, robustness, fail-safe, penetration, and adversarial-test results.
- Logging design, audit interfaces, and retention capabilities.
- Version history, dependency inventory, change-notification commitments, and incident history.
- GPAI documentation where a general-purpose model is integrated.
- Security certifications, data-processing terms, service levels, and regulatory cooperation clauses.

Sources

- Consolidated EU Artificial Intelligence Act
- Regulation (EU) 2026/1744, Digital Omnibus on AI
- European Commission AI Act overview and timeline
- Article 5 prohibited practices
- Annex III high-risk use areas
- Article 25 responsibilities along the AI value chain
- Article 26 deployer obligations
- Article 27 Fundamental Rights Impact Assessment
- Article 50 transparency guidance
- EU AI literacy guidance